

Weakly Quantum-Inspired Navigation: A Pilot-Wave World Model That Finds Routes a Local Greedy Policy Cannot

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Contents

Weakly Quantum-Inspired Navigation: A Pilot-Wave World Model That Finds Routes a Local Greedy Policy Cannot	1
Abstract	1
1. Introduction and motivation	2
2. World model and agents	2
3. Experimental design	3
4. Results	4
5. Applications	7
6. Discussion	9
7. Limitations	9
8. Conclusion	9
Data and code availability	10

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A research prototype report · QIWM (Quantum-Inspired World Model)

Code, data & scripts: <https://github.com/BrutchsamaJeanLouis/quantum-inspired-sensor-navigation> ·
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Scope statement. This is a 10-week research prototype, not a production system and not a claim that these dynamics are quantum mechanics. We build a classical grid world, add a *quantum-inspired* sensing channel (a Bohmian-style pilot wave that globally leaks around obstacles) plus a Penrose-style observation collapse, and ask a falsifiable question: *does that channel help a swarm of agents navigate, versus a classical local-greedy baseline?* Everything is “inspired by,” deliberately so. The honest answer, after instrument forensics, is **yes in four of five scenarios, with a specific mechanism (global route discovery) and a specific cost (coupling must be weak in dynamic fields, and directness is traded for reachability).**

Abstract

We test the null hypothesis that quantum-inspired dynamics provide *no* advantage over classical agent navigation, in a grid world model with energy sources, obstacles, and a toroidal boundary. The “quantum” agent reads a *guidance field* = the classical potential **plus** a diffusing, globally-leaked pilot-wave channel, and makes decisions by Boltzmann sampling over a 5×5 patch; a Penrose-inspired collapse (local coherence decay + pilot-wave sharpening) fires on each observation. The classical baseline is deterministic argmax.

After fixing a baseline bug found by reproducibility forensics (the v1 “classical” arm was actually the quantum sampler in argmax mode — 70–98% survival vs the true deterministic 0–45%), the calibrated instrument gives: the null is rejected in **4 of 5 scenarios** (tunnel 0%→100% alive, $p \leq 2.4e-5$; maze 0%→100%, $p \leq 1.6e-5$; single-source 20%→100%, $p \leq 1.6e-5$; default 45%→90%, $p \approx 7.6e-4$), with the moving scenario showing a **monotone coupling-liability curve** (advantage at $q=0.05$ – 0.15 , decaying to baseline by $q \geq 0.2$; $n=50$, $p < 5e-5$ at $q=0.05$). A crossing taxonomy shows the mechanism is **global route discovery via the leaked pilot wave, not wall penetration** — quantum agents wrap the torus seam (~ 2.5 -step latency), they never jump the wall; deterministic local-greedy classical is structurally unable to find that route (its gradient always points at the wall). The tradeoff is *route for the same price*: adding ϵ -greedy noise to the classical arm recovers up to 83.5% tunnel survival but at 5 – $12\times$ first-crossing latency, while the quantum arm stays at 100% and ~ 2.4 – 2.8 latency. The IIT-inspired Φ metric was re-scoped to an agent-in-the-loop, agent-local coupling measure; it is **not** $1.5\times$ higher under quantum (ratio 0.874, direction reversed), so that specific threshold is formally retracted while the re-scoped Φ is retained as a clean coupling diagnostic. We frame the result as an honest, instrumented null- rejection with a known mechanism, a known cost, and a documented retraction — not a silent win.

Keywords: world models, pilot-wave dynamics, IIT-inspired coherence, agent navigation, ablation, reproducibility.

1. Introduction and motivation

Most agent navigation research assumes the policy can see what it needs: a gradient, a reward, or a map. We want to know something smaller and more awkward: **if the only “sensing” an agent gets is a field that leaks around obstacles the way a wave does, does that help it reach things a local policy cannot see how to reach?**

This is a toy, deliberately. A 128×128 grid, 20 agents, energy sources ($1/r$ attraction) and repulsive obstacles, a toroidal boundary, 500 steps. Two arms:

- **Classical.** The agent reads the classical potential field locally (5×5 patch) and moves by deterministic argmax toward the steepest descent.
- **Quantum-inspired.** The agent reads a *guidance field* = potential + a coupling-weighted, normalised pilot wave. The pilot wave is a diffusing field (`np.roll-based` Laplacian) that does not respect the obstacle potential the way the local gradient does — it *leaks around* walls. The agent decides by Boltzmann sampling over the patch (temperature 1.0). Each observation triggers a collapse: local coherence decays and the local pilot wave is sharpened (Penrose-style).

The question is falsifiable: if the quantum arm does not beat the classical arm on survival, crossing, or route completion across a set of scenarios, the null stands. We build the instrument, run a 400-run ablation, and — because the first instrument was wrong — we document the forensics that caught it. That forensics is arguably the most reusable part of the project.

Why “quantum-inspired” and not “quantum”. The pilot wave is a classical diffusion field we *label* with the pilot-wave name because its useful property — global leakage around a barrier that a local gradient cannot see — is the property we care about. We do not implement amplitudes, superposition, or measurement in the quantum-mechanical sense. We are bridge-building, and we say so.

2. World model and agents

2.1 The world

ToyWorld is a `size×size` (default 128) grid with a scalar potential field. Energy sources are $(x, y, \text{strength}) \rightarrow$ attractive $1/r$ potential; obstacles are $(x, y, \text{radius}) \rightarrow$ repulsive potential ($-1e6$ inside the core in the tunnel scenario, i.e. an effective wall). The boundary is a **torus** by default (wrap-around), which is what makes the “seam shortcut” geometry (§4.2) possible. A `closed` boundary option (clamp instead of wrap) supports maze-style geometries. The potential is recomputed on demand; the classical field is *deterministic and, in static scenarios, static* — a fact that matters for the Φ result in §4.9.

2.2 The quantum-inspired channel

QuantumInspiredWorld extends the world with three arrays: a **pilot wave** (diffusing), a **coherence** field, and a **guidance field**. The pilot wave integrates a diffusion update (rate `diffusion_rate`, gain 3000) and is *normalised* before it is mixed in. The guidance field the agent actually perceives is

```
guidance = potential + quantum_coupling * 3000 * normalize(pilot_wave)
```

so at `quantum_coupling = 0` the perceived field *is* the classical potential (the honest baseline), and the coupling weight is the single knob that turns the channel on. Two properties matter, and both follow directly from this:

1. **Global leakage.** The pilot wave is a field that diffuses across the whole grid; the $-1e6$ wall cell is a spike in the *potential*, not a hard boundary in the *wave*. So the wave on the far side of a wall is non-zero, and the far side becomes *attractive* in the guidance field even where the local potential says “wall.” This is the route-discovery mechanism.
2. **Staleness under change.** The wave diffuses on a timescale set by `diffusion_rate`; when a source *moves*, the wave lags the source. A strong coupling glues the agent to a stale wave. This is the coupling-liability mechanism (§4.4).

2.3 The collapse

Each agent observation calls `collapse_field(x, y, radius=3)`: coherence in a local disk decays (`collapse_coherence_decay=0`), and the local pilot wave is amplified (`collapse_wave_amplify=2.0`) then renormalised. Operationally this sharpens the field where the agent has “looked.” We treat it as the *controller* in the sensor/controller split of §4.3, and ablate it directly.

2.4 Decision rules

Within a `sense_radius=2` (5×5) patch, the agent computes a value per cell (quantum arm: potential-based value + a local-coherence bonus; classical arm: the same value without the bonus). **Quantum** arm samples the move with Boltzmann sampling ($\exp(\text{value}/1.0)$). **Classical** arm takes `argmax`. In v1 the “classical” arm *also* sampled (a bug, §3.4); in v2 it is a genuine deterministic `argmax`. A small `sense_bonus` (0.05) rewards moving toward higher-coherence cells.

2.5 The IIT-inspired Φ metric

Two Φ measures exist. The original `compute_phi` takes a fixed region, deep-copies the world, forward-simulates a few steps *with no agents*, and returns $\max(0, \text{corr}(t0, t1) - \text{corr}(\text{boundary}, t1))$ — a self-predictability-vs-external-drive score. As we found (§4.9), with no agents in the loop and a region the agents never occupy, this measures **field** self-organization. The re-scoped `compute_phi_agents` runs the forward sim **with the whole swarm stepping** (sense/decide/move/collapse) and probes a $(2r+1)^2$ window around each agent’s *current* position, on the guidance field the agent perceives, averaged over agents. That is an agent–world **coupling** measure. Both are “IIT-inspired” in that high Φ = internally self-organized rather than externally driven; neither computes integrated information in the formal IIT sense.

3. Experimental design

3.1 Scenarios

Five built-in scenarios, each a fixed layout (128×128, 20 agents, 500 steps, corner spawn): **default** (one central source, three obstacle clusters), **tunnel** (a gapless wall column at $x=64$ separating the swarm from its only source), **maze** (hand-crafted wall loops with a goal source at (110,110)), **single_source** (one source, no obstacles), **moving** (three sources on circular orbits, sources move every 60 steps). Each scenario is run at coupling $q \in \{0.0, 0.1, 0.3, 0.5\} \times \text{collapse} \in \{\text{ON}, \text{OFF}\} \times 10 \text{ seeds} = 400 \text{ runs}$.

3.2 Ablations

The two ablation axes are **coupling** (channel strength) and **collapse** (observation controller ON/OFF). The coupling axis isolates “how much of the leaked wave the agent trusts”; the collapse axis isolates “does the observation gate help.” The ϵ -greedy classical control (§4.8) and the $n=50$ moving sweep (§4.4) are dedicated follow-up audits, not part of the 400-run grid.

3.3 Statistics

Per (scenario, q , collapse): alive rate, survival steps, final energy, barrier crossings, mean $\pm$ std across 10 seeds, with **bootstrap 95% CIs** (40k resamples). Pairwise quantum-vs-classical comparisons use a **two-sided Mann-Whitney U** test ($\alpha=0.05$). The $n=50$ moving sweep uses the same U test at each coupling level. We report p-values and CIs, not just means, and we report *insufficient evidence* as a distinct verdict from *rejected*.

3.4 Instrument forensics (v1 $\rightarrow$ v2) — the reusable part

The first instrument (v1) produced a “classical” baseline with 70–98% survival, which made the quantum arm look unremarkable. Reproducibility forensics found the bug: the v1 “classical” arm ($q=0.0$) was *not* a classical policy — it was the **quantum sampler in argmax mode**, i.e. it had already been handed the leaked pilot wave and was just taking the max instead of sampling. It was a quantum agent wearing a classical costume. The v2 fixes:

1. **True classical baseline:** $q=0.0$ now perceives *only* the classical potential (the guidance field reduces to the potential) and decides by deterministic argmax over the 5×5 patch.
2. **Deterministic tie-breaking** (stable argmin with a fixed index order), so a “seed” no longer perturbs a policy that should be seed-invariant.
3. **Pilot-wave prewarm** (2000 diffusion steps before agents spawn), so the channel is at steady state rather than transient for the first steps.
4. **Collapse ON by default** (the controller is part of the system, not an option).
5. A **2000-step prewarm + fixed RNG order** makes the 400-run dataset **byte-identical on re-run** (verified: `data/ablation_results_v2.csv` reproduces exactly).

With the calibrated instrument, the true classical survival is **0–45%** (not 70–98%), which is what makes the quantum advantage real and measurable. The lesson: **an ablation is only as good as its baseline, and a baseline that shares the treatment’s sensing channel is not a baseline.**

3.5 Reproducibility

The full 400-run dataset reproduces byte-identically in ~ 71 s on a 2014 dual-core (i3-4100), CPU-only, no GPU. A GitHub Actions CI runs the unit tests plus a 15 s headless UI smoke and a 25 s export smoke. All headline numbers are reproducible from the committed CSVs and the scripts in `examples/`.

4. Results

4.1 Survival: the null is rejected in 4 of 5 scenarios

The null — *quantum-inspired dynamics provide no advantage over classical* — is rejected (two-sided Mann-Whitney U, $\alpha=0.05$, $n=20$ per row) in:

scenario	classical ($q=0$)	quantum (best q)	U-test	verdict
tunnel	0% alive, 0 crossings	100% alive, 20/20 crossings	$p \leq 2.4e-5$	decisive
maze	0% alive	100% alive at every $q > 0$	$p \leq 1.6e-5$	decisive
single_source	20% alive	100% alive	$p \leq 1.6e-5$	decisive
default	45% alive	90% at $q=0.3/0.5$	$p \approx 7.6e-4$	significant

scenario	classical (q=0)	quantum (best q)	U-test	verdict
moving	45% alive	68% at q=0.05 (n=50)	p<5e-5	significant (low q only)

Bootstrap CIs (40k resamples) confirm: tunnel q=0.3 alive 1.000 [1.000, 1.000] vs classical 0.000 [0.000, 0.000]; default q=0.3 alive 0.900 [0.700, 1.000] vs 0.450 [0.450, 0.450]. The default scenario is *insufficient evidence* at q=0.1 (CI crosses zero) and becomes significant at q=0.3/0.5 — we report that gradient rather than picking the best cell.

4.2 Mechanism: global route discovery, not wall penetration

The tunnel is the cleanest geometry, and a **crossing taxonomy** (first crossing per agent, classified as wall-jump / wall walk-through / seam-wrap, 10 seeds each) tells the mechanism:

arm	wall jumps (63 65)	wall walk-throughs (via 64)	seam wraps	1st-crossing latency
classical ($\epsilon=0$)	0	0	0	—
ϵ -greedy 0.1	4	124	20	17.4
ϵ -greedy 0.3	15	157	24	5.2
quantum q=0.1 ($\epsilon=0$)	0	0	169	2.8
quantum q=0.5 ($\epsilon=0$)	0	0	200	2.4

Quantum agents never jump the wall (0 wall-jumps and 0 walk-throughs across all 100 quantum agents — Boltzmann sampling assigns ~zero mass to the $-1e6$ cell). They **wrap the torus seam** (~2.5 steps after reaching it). The globally leaked pilot wave makes the far side attractive, and Boltzmann sampling discovers the route that a local greedy policy *structurally* cannot find: the classical gradient always points at the wall, never at the seam ($x=0/127$). A maze route audit confirms the same mechanism: agents reach the maze goal by **wrapping the edge (1.00 fraction)**, not through the hand-crafted gaps (0.00 for quantum, 0.02 for ϵ -classical). The quantum-specific delta is **completeness** (20/20 reach, 100% survival) vs ϵ -classical (11.4/20 reach) — the quantum channel makes the exploration exhaustive.

Retraction (honest): the v2-era claim “quantum crosses the wall while classical cannot” was wrong. The defensible claims are: (1) deterministic local-greedy classical never reaches the far side (structural); (2) stochastic classical (ϵ -greedy) *does* find the route, so the tunnel separates **determinism vs exploration**, not quantum vs classical per se; (3) the quantum channel shortens first-crossing latency (2.4–2.8 vs 5.2–17.4) and survives ~100%.

4.3 Collapse: the difference between survival and death in dynamic worlds

The collapse ablation splits cleanly by geometry:

- **Static geometries (tunnel, single_source):** the pilot-wave channel *alone* suffices — full advantage with collapse **OFF** (tunnel collapse-OFF q=0.3: 100% alive, 20/20 crossings).
- **Trapped/dynamic geometries (default, moving):** quantum with collapse **OFF** does *worse than classical* (default 45–48% $\approx$ baseline; moving 0% vs 45%). The advantage appears **only with collapse ON** (default $\rightarrow$ 90%; moving 39–97% at low q).

Reading: **the pilot wave is the sensor; the collapse/coherence gate is the controller.** The controller switches between Boltzmann exploration and greedy exploitation where local coherence has been measured. In static fields the sensor alone wins; against moving fields and trap geometries, sensing *without* the collapse-driven decision switch is actively fatal.

4.4 Dynamic fields: a coupling-liability curve (n=50)

The moving scenario (sources orbit, recompute every 60 steps) at **n=50** (8 coupling levels $\times$ 50 seeds, collapse ON) shows a real, **monotone** effect — a significant quantum advantage at low q that decays to the classical baseline as q grows:

q	0.0	0.05	0.1	0.15	0.2	0.3	0.4	0.5
alive (n=50)	0.450	0.679	0.653	0.621	0.569	0.528	0.451	0.435
p vs q=0	—	<5e-5	8e-4	9e-3	0.14	0.46	0.46	0.26

Mechanistic read: the classical field updates *instantly* when a source moves; the pilot wave **lags**. A *weak* channel adds non-local sensing (advantage); a *strong* channel glues the agent to a stale wave (liability). There is an **interior optimum at low coupling** for dynamic fields, while the static scenarios saturate by $q=0.1$. Operating point: **weak coupling for dynamic fields, anything above for statics**. (Earlier $n=10$ runs were underpowered/noise; the $n=50$ sweep is the one we cite.)

4.5 Anticipation: lag reduction, not foreknowledge

A dedicated lead-horizon metric ($H=15$ steps; index = mean($d_{\text{now}} - d_{\text{lead}}$), positive = positioned toward where the source is *heading*; $n=10$):

q	index (mean $\pm$ std)	frac steps ahead
0.0	-4.09 ± 0.00	0.000
0.1	-2.58 ± 2.70	0.071
0.3	-1.39 ± 2.94	0.103

Honest null as an anticipation claim: no condition shows *positive* anticipation — every agent, including quantum, sits on average *closer* to the source’s current than future position (the diffusive field carries *past* information, not future). The quantum channel monotonically **reduces the lag** ($-4.09 \rightarrow -2.58 \rightarrow -1.39$) and forward positioning appears (frac $0 \rightarrow 0.07 \rightarrow 0.10$). The advantage is *reduced tracking error in a shifting field*, not foresight.

4.6 Efficiency: directness is traded for reachability

Path efficiency = (spawn $\rightarrow$ final distance) / (steps taken), on trajectories that reach a source. **Classical is more direct; quantum reaches more.** Default: classical 1.0207 (9/10 reach) vs quantum 0.5055 (20/20 reach). Maze: classical **0.0000** (0/20 reach) vs quantum 0.8940 (20/20). The readme’s path_efficiency 1.15 $\times$ threshold is *not* met — and that is the right result, because the quantum arm’s win is **reachability** (it gets there at all), while the classical arm’s “efficiency” is partly the efficiency of *giving up* (it never reaches, so its short paths are cheap). We report the tradeoff, not the artifact.

4.7 Energy cost

Per-run mean net energy (final – initial): classical always dies at 0 net (initial 300, harvest $0 \rightarrow$ but net is measured over the run). Quantum is *slightly cheaper per surviving run* in the tunnel (-0.266 vs -0.446) and *more expensive* in the maze ($+0.08$ vs $+1.54$ classical — classical dies at 0 energy; the $+1.54$ is the mean over the few that harvest). The energy budget (examples/energy_budget.py) exists to keep “quantum is more expensive” an *measured* claim, not an assumption. The honest summary: the channel is cheap enough that it does not eat the survival advantage, and the per-run cost is dominated by *whether the agent survives to harvest*, not by the sensing itself.

4.8 Noise robustness: route for the same price (ϵ -sweep)

A 5-level ϵ -greedy classical control (7 arms $\times$ 10 seeds, tunnel) shows the tradeoff precisely:

arm	alive	1st-crossing latency (median)	wall walk-throughs/agent	seam wraps/agent
classical $\epsilon=0$	0.00	—	0.0	0.0
$\epsilon=0.05$	0.480	21.5	9.7	2.0
$\epsilon=0.1$	0.665	13.8	12.0	2.3

arm	alive	1st-crossing latency (median)	wall walk-throughs/agent	seam wraps/agent
$\epsilon=0.2$	0.790	11.2	15.8	2.2
$\epsilon=0.3$	0.835	7.0	14.9	2.8
quantum $q=0.3$ ($\epsilon=0$)	1.000	2.0	0.0	20.0
quantum $q=0.3$ + $\epsilon=0.1$	1.000	2.0	0.0	20.0

Noise *helps* the classical arm cross (monotone $0 \rightarrow 48 \rightarrow 66.5 \rightarrow 79 \rightarrow 83.5\%$ alive as ϵ rises $0.05 \rightarrow 0.3$) but at a steep latency cost (quantum median 2.0 vs 7.0–21.5 for the ϵ -arms, **3–10 $\times$**), and the ϵ -taxonomy shifts (wall walk-throughs $0 \rightarrow \sim 16/\text{agent}$; seam stays flat $\sim 2/\text{agent}$; true 2-cell jumps $0.3 \rightarrow 1.6/\text{agent}$). **Parity (100% alive) is never reached** at any $\epsilon \leq 0.3$. Adding ϵ *on top of* the quantum arm is **benign** (100% alive, 20/20 seam wraps, median latency 2.0) — the quantum channel is not destroyed by exploration noise. Framing: the quantum channel is **a route for the same price**, not parity with a noisier classical policy.

4.9 Φ re-scope: an agent–world coupling diagnostic

The original `compute_phi` (agent-less forward sim on a fixed central region the agents never occupy) measured **field** self-organization $\rightarrow$ ratio **1.000** across conditions (an instrument artifact, not a finding). We re-scoped it to `compute_phi_agents`: the forward sim runs **with the swarm stepping** (sense/decide/move/collapse) and probes a 9×9 window around each agent’s current position, on the guidance field the agent perceives, averaged over agents — a **coupling** measure. Defined before the number was read ($n=10$, default scenario, collapse ON):

config	phi_agent (mean $\pm$ std)	min–max
classical $q=0$	1.0000 ± 0.0000	1.0000–1.0000
quantum $q=0.3$	0.8740 ± 0.0125	0.8546–0.8883

The readme threshold `quantum_ $\Phi$  > 1.5  $\times$  classical_ $\Phi$` is **NOT MET** (ratio 0.874) and the **direction is reversed** (quantum lower). Moving scenario: classical 1.0000, quantum 0.4630 ± 0.1965 — same direction. The direction is consistent across **three** probe designs (field-only 1.000, swarm-centroid quantum-lower, agent-local quantum-lower).

Instrument property (documented, not a finding): classical is pinned at 1.0000 in every scenario because its perceived field is the deterministic potential, locally *static* over the 5-step horizon $\rightarrow t_0 t_1 \rightarrow$ self-predictability 1.0 (a ceiling). Quantum is < 1.0 because its perceived field adds the diffusive pilot-wave channel, which is *evolving* and therefore less self-predictable over 5 steps. So the *magnitude* (0.874) is partly a ceiling effect; the *direction* is the robust signal.

Decision: we **formally retract** the specific “quantum Φ is $1.5 \times$ classical Φ ” threshold as stated, and **retain** the re-scoped Φ as a clean agent–world- coupling diagnostic. The paper carries this as an honest negative with a fixed instrument and a consistent direction — not a silent null.

5. Applications

The practical value of the result is not the word “quantum” in the title — it is that the finding **replaces a fantasy with a mechanism**, and hands you the operating map. A “clean win” (“quantum is $1.5 \times$ better, ship it”) would have been *less* useful: it would not tell you when the channel breaks (it does, in the dynamic/moving case at high coupling). “Route discovery, not wall penetration” is *buildable* — a specific architecture you bolt onto an existing local policy that measurably reaches more — and the ablation tells you exactly how to deploy it (weak coupling, on the gate). Ranked most $\rightarrow$ least impactful, here is what this everything enables, and the deliverable each one becomes.

5.1 A drop-in exploration channel for local-greedy navigation agents

Most impactful — directly productizable. Every agent that gets stuck in local minima today — warehouse AMRs, game NPCs, multi-agent swarms, network routing, protein/sequence search, any gradient-descent-style policy — can add this stack *on top of the existing greedy policy*, not instead of it: a cheap **globally-leaked sensing channel** (here, pilot-wave diffusion) + **Boltzmann sampling** + a **decision gate**, with the greedy policy as the fallback. The finding tells you the correct operating point — *weak coupling, always on the gate* — so it is shipped without the v1 “full-strength” bug. **Deliverable:** a library / agent mode (“greedy policy + leaky-sensor exploration”) whose spec is the ablation CSV; a before/after on any local-greedy nav stack shows the reachability gain (maze: 0/20 → 20/20 arrivals).

5.2 A reusable “is your baseline really a baseline?” eval method

The paper’s most reusable piece. The instrument forensics caught a control that shared the treatment’s sensing channel (the v1 “classical” arm was the quantum sampler in argmax mode). The general rule — *an ablation is only as good as its baseline; a baseline that shares the treatment’s channel is not a baseline* — applies to any A/B or feature-ablation claim in ML. **Deliverable:** a test/audit harness + checklist (deterministic baseline, deterministic tie-breaking, prewarm, fixed RNG order) that catches “the control is wearing the treatment’s costume,” with the v1→v2 forensics as the worked example.

5.3 A sensor-vs-controller decomposition rule

The cleanest causal statement in the project: **the leaky channel is the sensor; the collapse is the controller; they are not interchangeable** — sensing without the collapse-driven decision switch is *worse than classical* in dynamic/trapped worlds (moving: 0% vs 45%). This is a design rule for any sensing/planning split (robot perception+control, ranking+recsys, monitor+reactor): *when you add a new sensing channel, co-design its decision gate, or it backfires*. **Deliverable:** a one-page design pattern with the moving-scenario numbers as the proof.

5.4 “Completeness, not directness” as a product axis

The path-efficiency result (quantum ratio $0.495 < 1.15$, i.e. *worse* on directness) reframes the “quantum is less efficient” objection as *“less direct, but actually arrives”* (maze: classical 0/20 at any efficiency, quantum 20/20). For search/retrieval/recommender products, “recall over precision, exhaustive over fast” is a real, defensible positioning axis. **Deliverable:** copy + a metric (“reachability @ equal cost”) for any product where missing the reachable target matters more than route length.

5.5 A coupling auto-tuning rule

The $n=50$ moving sweep is a monotone “*weak coupling wins, strong coupling is a liability*” curve (a moving target → the sensor lags). **Deliverable:** a controller that **down-regulates coupling in dynamic fields** — a concrete, measurable policy rather than a guess at a fixed hyperparameter.

5.6 Honest-nulls as a “when NOT to use” boundary

The ϵ -sweep (noise alone reaches 83.5%, never 100%) and the closed-boundary audit (remove the seam, *nobody* tunnels) give clear negative boundaries: no impenetrable-barrier penetration; plain stochastic noise is a cheap *partial* substitute. **Deliverable:** a one-page applicability boundary — *use when occluded/contractible routes exist; do not expect wall-penetration or shorter paths*. This stops a team from over-buying the “quantum” framing.

5.7 A teaching / demo artifact

The interactive demo (scrubbable timeline, --shot headless primitives, A/B field + Φ overlay) is a *visual proof of the mechanism* — you can watch the leaked field light up the far side of a wall ~2 steps before the agent goes. **Deliverable:** a talk/teaching demo for “non-local route discovery without a global map.”

5.8 A reproducible, CPU-only benchmark

The whole thing is a tiny, **byte-reproducible** benchmark (128×128, 20 agents, one i3, ~71 s for 400 runs) for the question “does global sensing beat local greedy here?” **Deliverable:** a benchmark harness others can plug new scenarios or policies into.

6. Discussion

Three claims survive the instrument forensics and the ablations:

1. **A globally-leaked sensing channel lets a swarm reach things a local greedy policy cannot structurally reach.** The mechanism is route discovery (wrap the seam / wrap the edge), not wall penetration, and it is robust across the tunnel and the maze. The quantum-specific delta is *completeness* (exhaustive exploration), because Boltzmann sampling over the leaked field assigns non-zero mass to routes the local gradient never points at.
2. **The channel is a sensor; the collapse is the controller.** The collapse ablation is the cleanest causal statement in the project: in dynamic/trapped geometries, sensing without the collapse-driven decision switch is *worse than classical*. The two are not interchangeable.
3. **The advantage has a known, cheap cost: coupling must be weak in dynamic fields, and directness is traded for reachability.** The n=50 moving sweep is the most honest number in the report (a monotone liability curve, not a “quantum wins everywhere” claim), and the path-efficiency result reframes the “quantum is less efficient” finding as “quantum is less direct but actually arrives.”

What we *retract* and what we *keep* is explicit: the wall-jumping mechanism (retracted → seam/edge route discovery), the “quantum Φ is 1.5×” threshold (retracted → re-scoped coupling diagnostic), and the v1 baseline (replaced by the calibrated deterministic classical). What we keep: the 4/5 null rejection, the sensor/controller split, and the coupling-liability curve.

Framing for honesty. This is a *negative-tilted positive*: a real, mechanism-explained advantage in 4/5 scenarios, a monotone cost curve in the fifth, and two honest retractions. We would rather report “advantage in 4/5 with a known cost and a documented retraction” than “advantage everywhere.” The instrument forensics (§3.4) is the piece we expect to be most reusable: a baseline that shares the treatment’s sensing channel is not a baseline.

7. Limitations

- **One world model, one grid, 20 agents, 500 steps.** Results are for *this* geometry family and scale. Generalization to larger/multi-scale worlds is future work, not a claim.
 - **The “quantum” is inspired, not quantum.** No amplitudes, superposition, or formal measurement; the pilot wave is a classical diffusion field we label with the name because of its leakage property.
 - **Φ is IIT-inspired, not integrated information.** Two proxy measures (self-predictability vs external drive); the re-scoped one is a coupling diagnostic, not a Φ computation in the IIT sense.
 - **n=10 for most of the 400-run grid** (n=50 only for the moving sweep). Some default-scenario contrasts are *insufficient evidence* at q=0.1; we report the gradient, not a point win.
 - **Static-field Φ ceiling.** Classical Φ is pinned at 1.0 by the static potential; the magnitude of the quantum Φ dip is partly a ceiling effect (the direction is the robust part).
 - **Single seed-order / RNG** for reproducibility; we did not re-derive the dataset under a different RNG (we re-ran it under the same one, byte-identical).
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8. Conclusion

A weakly quantum-inspired sensing channel — a pilot wave that leaks around obstacles, read through Boltzmann sampling, gated by a Penrose-style collapse — gives a swarm of grid agents a real, mechanism-explained navigation advantage over a deterministic local-greedy classical baseline in four of five scenarios (tunnel, maze, single-source, default) and a

coupling-dependent advantage in the fifth (moving). The mechanism is **global route discovery via the leaked sensor**, not wall penetration; the controller (collapse) is what makes sensing useful in dynamic fields; and the cost is a **weak-coupling operating point in dynamic fields** plus **directness traded for reachability**. After instrument forensics caught a costume-wearing classical baseline, the honest report is a 4/5 null-rejection with a known mechanism, a known cost, a monotone liability curve, and two explicit retractions (wall-jumping; the $1.5 \times \Phi$ threshold). That is, in our view, a more useful result than a clean win would have been.

Data and code availability

- **Repository:** <https://github.com/BrutchsamaJeanLouis/quantum-inspired-sensor-navigation> (world model, agents, Φ , ablation harness, all scripts, tests, and this report).
- **Paper DOI (Zenodo):** <https://doi.org/10.5281/zenodo.22963620> (10.5281/zenodo.22963620)
- **Dataset** (all under data/): ablation_results_v2.csv (400 runs, byte-reproducible), q_sweep_moving_n50.csv, epsilon_sweep_tunnel.csv, phi_rescope.csv, maze_route_audit.csv, path_efficiency_*.csv, energy_budget.csv, moving_anticipation_n10.csv, phi_audit.csv, tunnel_closed_audit.csv.
- **Scripts:** examples/ (one script per table above); src/core/ (world, agents, coherence, experiments).
- **Reproduce:** `pip install -r requirements.txt && python examples/ablation_v2.py`, then `python examples/analyze_results.py`. **Unit tests:** `pytest tests -q` (120 passed). **CI:** `.github/workflows/ci.yml`.
- All headline numbers trace to a committed CSV and a named script; the re-scoped Φ audit is `python examples/phi_rescope.py`.